

Assignment 14

1. Create an ATM program handling invalid withdrawal amount using try-catch

Code:

```
import java.util.Scanner;
```

```
public class ATMException {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        double balance = 10000;  
  
        try {  
            System.out.print("Enter withdrawal amount: ");  
            double amount = sc.nextDouble();  
  
            if (amount <= 0) {  
                throw new IllegalArgumentException("Withdrawal amount must be  
greater than zero.");  
            }  
  
            if (amount > balance) {  
                throw new IllegalArgumentException("Insufficient balance.");  
            }  
  
            balance = balance - amount;  
  
            System.out.println("Withdrawal successful.");  
            System.out.println("Remaining balance: " + balance);  
  
        } catch (IllegalArgumentException e) {  
            System.out.println("Error: " + e.getMessage());  
        } catch (Exception e) {  
            System.out.println("Invalid input.");  
        }  
  
        sc.close();  
    }  
}
```

```
}
```

Output:

```
Enter withdrawal amount: -500
Error: Withdrawal amount must be greater than zero.
```

For insufficient balance:

```
Enter withdrawal amount: 15000
Error: Insufficient balance.
```

2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.

Code:

```
import java.util.Scanner;
```

```
public class OnlineShoppingException {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter product quantity: ");
            int quantity = sc.nextInt();

            if (quantity <= 0) {
                throw new IllegalArgumentException(
                    "Quantity must be greater than zero."
                );
            }

            System.out.println("Product quantity accepted: " + quantity);
            System.out.println("Order placed successfully.");

        } catch (IllegalArgumentException e) {
            System.out.println("Error: " + e.getMessage());
        } catch (Exception e) {
            System.out.println("Invalid input. Please enter a number.");
        }

        sc.close();
    }
}
```

```
}  
}
```

Output:

```
Enter product quantity: 0  
Error: Quantity must be greater than zero.
```

Valid input:

```
Enter product quantity: 3  
Product quantity accepted: 3  
Order placed successfully.
```